Accident Hotspot Prediction Using Random Forest and Spatial Clustering

MACHINE LEARNING

UNDER THE GUIDENCE OF
JYOSHNA

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Problem Statement & Objectives

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Road accidents cause significant loss of life, injuries and economic damage. Accident locations are influenced by several factors such as traffic density, weather, road conditions, time of day and human errors. Traditional accident analysis mainly depends on historical reports and manual investigation, making it difficult to identify hidden patterns and future high-risk locations.

Objectives

- Predict accident risk using Random Forest
- Identify accident-prone geographical locations
- Discover accident hotspots using spatial clustering
- Analyze factors contributing to accidents
- Visualize high-risk areas on maps
- Support traffic authorities in preventive decision-making

Dataset & EDA Dataset

Indian Road Accident Dataset

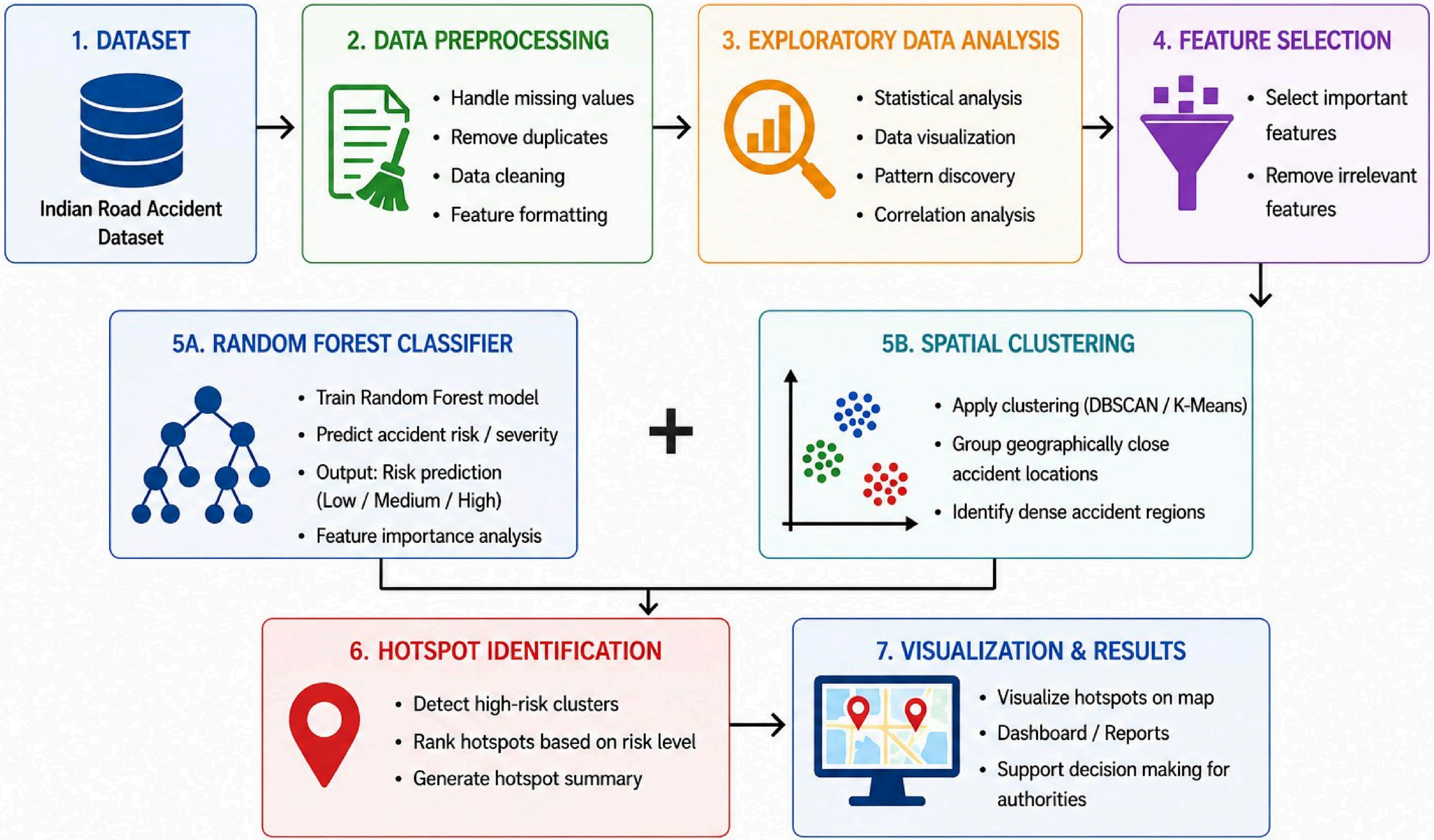
- Latitude & Longitude
- City & State
- Hour / Day of Week
- Road Type
- Weather
- Traffic Density
- Vehicles Involved
- Casualties
- Accident Severity
- Risk Score

EDA Performed

- Accident severity distribution
- Accidents by city and state
- Accident causes
- Weather and traffic analysis
- Accident frequency by hour
- Risk-score distribution
- Correlation analysis
- Geographical accident distribution

PROPOSED METHODOLOGY

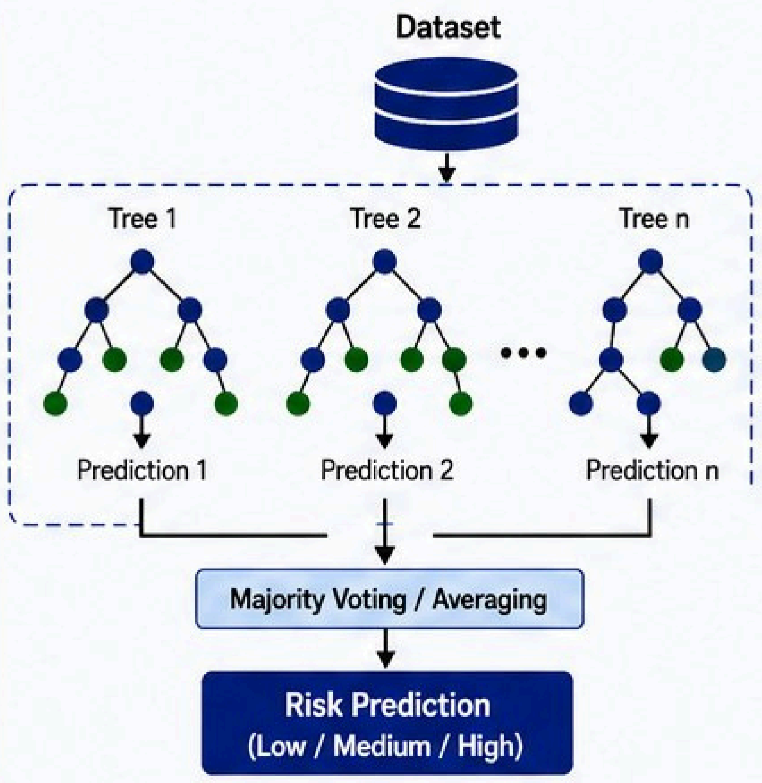
The proposed system follows a structured workflow to predict accident risk and identify accident hotspots.



TECHNICAL APPROACH & TOOLS

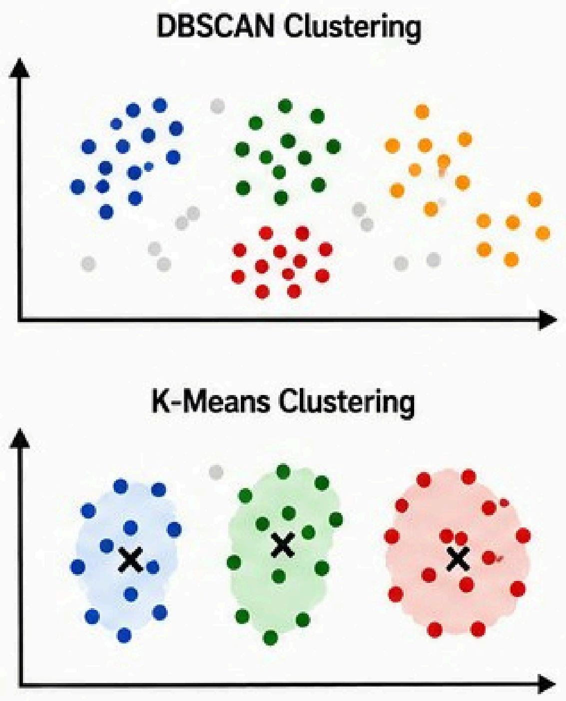
The system integrates Machine Learning (Random Forest) with Spatial Clustering techniques to predict accident risk and discover accident hotspots.

1. RANDOM FOREST CLASSIFIER



- Handles high dimensional data and complex relationships
- High accuracy and robust to overfitting
- Provides feature importance
- Predicts accident risk / severity levels

2. SPATIAL CLUSTERING (DBSCAN / K-MEANS)



- Uses latitude and longitude of accident locations
- Groups nearby accident points into clusters
- Identifies dense regions as accident hotspots
- Helps in geographic visualization and analysis

3. TOOLS & LIBRARIES USED

- **Python** Programming Language
- **Pandas** Data Handling & Processing
- **NumPy** Numerical Computing
- **Scikit-learn** Machine Learning (Random Forest)
- **DBSCAN / K-Means** Spatial Clustering
- **GeoPandas** Geospatial Data Processing
- **Matplotlib & Seaborn** Data Visualization
- **Folium** Interactive Map Visualization

4. TECHNOLOGY STACK

Language	Python 3.10+
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn (Random Forest)
Clustering	DBSCAN / K-Means
Spatial Processing	GeoPandas
Visualization	Matplotlib, Seaborn, Folium
Development Environment	Jupyter Notebook / VS Code

Thank You

For Your Attention

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